

# Which Regression Line Fits?

Unit 5 · Topic 5.5 · TODAY'S PROBLEM

Suppose a solar-farm analyst uses sunshine hours  $x$  to predict daily generation  $y$  in megawatt-hours (MWh). The table summarizes 12 observed days.

**Quinn:** “Using  $b = r(s_y/s_x)$  gives  $\hat{y} = 5.6 + 3.2x$ . It passes through (7, 28), so it must be the LSRL. It explains 80 percent of the variation.”

**Rowan:** “Using  $b = r(s_x/s_y)$  gives  $\hat{y} = 26.6 + 0.2x$ . It also passes through (7, 28), so it is equally good. Explained variation is 64 percent.”

## Solar-farm summary (12 observed days)

| $x$ range | $\bar{x}$ | $\bar{y}$ | $s_x$ | $s_y$ | $r$  |
|-----------|-----------|-----------|-------|-------|------|
| 4–10 h    | 7 h       | 28 MWh    | 1.5 h | 6 MWh | 0.80 |

### FIRST TAKE (5 min, on your sheet)

The two lines agree at 7 sunshine hours. Does that make them equally useful at other times? Give your first thought in one sentence.

- DOK 1** (a) State the mean point that every least-squares regression line must contain.
- DOK 2** (b) Calculate the regression slope and intercept, and  $r^2$ . Give the slope's units.
- DOK 3** (c) Which parts of Quinn's and Rowan's claims should a report keep? Use the slope formula to choose a line, explain why their shared mean point does not make both lines correct, and interpret the correct percent of variation explained. Write 3–4 sentences.

**Optional review:** [follow-along](#) (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.

